

# SHASHWAT MAHAJAN

+91-93569 97907 | [shashwatmahajan2@gmail.com](mailto:shashwatmahajan2@gmail.com) | [linkedin.com/in/shashwatmahajan2](https://www.linkedin.com/in/shashwatmahajan2) | [github.com/Shashwat-Mahajan](https://github.com/Shashwat-Mahajan)

## TECHNICAL SKILLS

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**Languages:** Python, Java, JavaScript (ES6+), C, SQL

**AI / ML:** LLMs, LangChain, RAG Pipelines, Prompt Engineering, Vector Embeddings, ChromaDB, HuggingFace, Groq API, XGBoost, Scikit-Learn, Pandas, Streamlit

**Web & Backend:** React.js, Next.js, Node.js, Express.js, FastAPI, MongoDB, PostgreSQL, WebSockets, REST APIs

**DevOps & Tools:** Docker, GitHub Actions, CI/CD Pipelines, Git, Linux, JWT

**Data Science:** Feature Engineering, SMOTE, RAGAS Evaluation, MLflow, DagsHub, Experiment Tracking

## EXPERIENCE

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### GenAI Engineer Intern — Persistent Systems

Generative AI Division | Jan 2026 – Present | Nagpur, IN

- Reduced vector retrieval p95 latency by ~60% through hybrid dense-sparse indexing and retrieval caching, directly improving response quality for 3 production GenAI features serving live end users.
- Took sole ownership of 2 AI-powered features end-to-end — prompt design, vector store tuning, deployment, and post-launch monitoring — both shipped on schedule with zero production regressions.
- Engineered sub-second RAG pipelines processing high-frequency data streams; designed for <300ms retrieval SLAs, applicable to real-time financial and transactional workloads.
- Optimised multi-system GenAI infrastructure for latency and accuracy; work ships directly to end users in a live, high-throughput environment.

## EDUCATION

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Yeshwantrao Chavan College of Engineering, Nagpur | **B.Tech Computer Science Engineering** | CGPA: 8.00 / 10 | Expected June 2027

*Relevant Coursework:* Machine Learning · Data Structures & Algorithms · Database Systems

## PROJECTS

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### Financial Analyst RAG System [\[GitHub\]](#)

Python | LangChain | ChromaDB | Groq API | HuggingFace Embeddings | Streamlit | AWS S3 | GitHub Actions | RAGAS

- Deployed a production-grade RAG pipeline ingesting live financial news from 10+ sources, delivering sub-2s retrieval for real-time multi-company portfolio queries with full source citation — directly comparable to the data-pipeline demands of a trading platform.
- Automated continuous ingestion via GitHub Actions scheduled workflows; enforced RBAC to restrict query access by user tier; persisted ChromaDB vector store to AWS S3 for durable, cost-efficient retrieval across sessions — zero manual intervention required.
- Implemented HuggingFace query-expansion for dynamic cross-company comparisons, LLM guardrails for safe output, and RAGAS evaluation framework for ongoing retrieval quality benchmarking.

### Smart Attendance System — Smart India Hackathon (SIH) | Team Lead, 6-Member Team

MongoDB | Express.js | React.js | Node.js | Supabase | PostgreSQL | WebSockets

- Led a 6-member cross-functional team at SIH National Hackathon: owned system architecture, sprint planning, task delegation, and live demo delivery — shipped a fully working product under national-competition pressure.
- Architected a real-time WebSocket synchronisation layer sustaining 100+ concurrent users with zero data loss under sustained load — the same low-latency, high-throughput concurrency model required for live stock market data pipelines.
- Designed rolling QR-based authentication to prevent replay attacks; enforced MFA and Supabase Row-Level Security (RLS) for strict per-user data isolation.

### Credit Risk Assessment System [\[GitHub\]](#)

React.js | FastAPI | XGBoost | Docker | GitHub Actions | CI/CD

- Architected a full-stack ML inference system for real-time credit risk predictions: React.js frontend → FastAPI backend → XGBoost classifier, serving sub-100ms results suitable for high-value financial transaction scoring at production traffic.
- Achieved 88% accuracy with 91% recall on the high-risk minority class (up from a 67% naive baseline) by training XGBoost on 10K+ records with SMOTE oversampling to correct severe class imbalance.
- Containerised with Docker and implemented a GitHub Actions CI/CD pipeline with automated test coverage on every commit, enabling zero-downtime deployments.

## HACKATHONS

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### DevOnboard — AI-Powered Repository Intelligence Platform | IBM Hackathon

Python | LangChain | RAG | Vector Embeddings | GitHub API | LLM

- Built an AI platform that ingests any GitHub repository via URL, parses its structure, README, and source code, and returns cited, context-aware answers to developer onboarding questions — eliminating hours of manual codebase exploration.
- Engineered a chunking-and-embedding retrieval pipeline using semantic vector search to deliver accurate Q&A; across large, multi-file codebases with full source attribution.
- Showcased at IBM Hackathon; solution compresses new-developer ramp-up time by automating codebase comprehension end-to-end.

### SITNOVATE — Real-Time Chat Engine | 24-Hour Hackathon

React | WebSocket | JWT

- Designed and shipped a JWT-secured, real-time WebSocket chat system entirely within 24 hours; scoped architecture under severe time constraints and delivered a live working demo to a judging panel.

## ACHIEVEMENTS & CERTIFICATIONS

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- **Competitive Programming:** LeetCode Rating 1538 — 400+ problems solved | CodeChef 2-Star | Primary Languages: Python & Java
- **Certifications:** Machine Learning Specialization — Codebasics.io | Full Stack Web Development (MERN Stack)